# Mediation analysis

Zhichao Jiang

Sun Yat-sen University

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# Causal mechanisms

- Scientists care about causal mechanisms, not just causal effects
- Randomized experiments often only determine whether the treatment causes changes in the outcome
- Not how and why the treatment affects the outcome

- Common criticism of experiments and statistics: black box view of causality

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- Not how and why the treatment affects the outcome

- Common criticism of experiments and statistics: black box view of causality

- Mediation analysis studies the extent to which an effect is mediated through a particular pathway and to which the effect of a treatment on the outcome operates directly

# Examples

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- Neighborhood poverty $\rightarrow$ school and peer environment $\rightarrow$ adolescent substance use
- Job training $\rightarrow$ job-search self-efficacy $\rightarrow$ mental health

# Difference and product methods

- Two methods commonly used in practice
- Product method for indirect effect:
  - regress $Y$ on $Z$, $M$ and $\mathbf{X} \to \widehat{\theta}_2$ (coef. for $M$)
  - regress $M$ on $Z$ and $X \to \widehat{\beta}_1$ (coef. for $Z$)
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  - estimator: $\widehat{\tau}_1 - \widehat{\theta}_1$
- How is the effect defined?
- What assumptions are needed to justify these methods?

# Robins-Greenland-Pearl nested potential outcomes

- Binary treatment $Z_i$
- Potential mediators and outcomes: $M_i(z)$ and $Y_i(z)$
- Potential outcomes under both $z$ and $m$: $Y_i(z, m)$

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- For example, $Y(1, M(0))$ is the hypothetical outcome if the unit received treatment 1 but its mediator were set at its natural value $M(0)$ without the treatment.
- Observed outcome: $Y_i = Y_i(Z_i, M_i(Z_i))$. If $Z_i = 1, M_i(1) = 0$, then $Y_i = Y_i(1, 0)$.

# Cross-world potential outcomes

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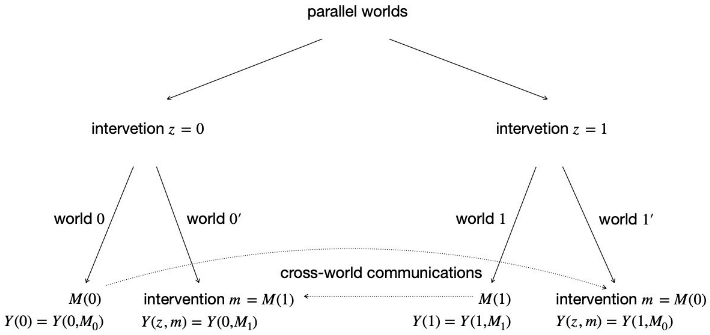
- $Y(1, M_i(0))$: interventions $Z = 1$ and $M = M(0)$ cannot simultaneously happen in any realized experiment
- We need to imagine the parallel worlds



# Metaphysics or science?

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- According to Popper (1963), a way to distinguish science and metaphysics is the falsifiability of the statements. That is, if a statement is not falsifiable based on any physical experiments or observations, then it is not a scientific but rather a metaphysical statement.
- A strict Popperian statistician would view mediation analysis as metaphysics

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- Natural indirect effect:
  $\text{NIE} = Y_i(1, M_i(1)) - Y_i(1, M_i(0)) = Y_i(1) - Y_i(1, M_i(0))$
- $\text{ACE} = \text{NDE} + \text{NIE}$

# Controlled direct and indirect effects

- Controlled direct effect: $Y_i(1,0) - Y_i(0,0)$
- Controlled indirect effect: $Y_i(1,1) - Y_i(1,0)$

# Controlled direct and indirect effects

- Controlled direct effect: $Y_i(1,0) - Y_i(0,0)$
- Controlled indirect effect: $Y_i(1,1) - Y_i(1,0)$

- Controlled indirect effect is not the effect of the treatment

$$\begin{tikzcd} & M \arrow[dr] & \\ Z \arrow[rr] & & Y \end{tikzcd}$$

$$\begin{aligned} \text{NIE} &= Y_i(1, M_i(1)) - Y_i(1, M_i(0)) = 0 \\ \text{CIE} &= Y_i(1,1) - Y_i(1,0) \neq 0 \end{aligned}$$

# Identification assumptions

- (A) No treatment-outcome confounding: $Z_i \perp\!\!\!\perp Y_i(z, m) \mid \mathbf{X}_i$
- (B) No mediator-outcome confounding: $M_i \perp\!\!\!\perp Y_i(z, m) \mid (Z_i, \mathbf{X}_i)$
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- (A)+(B): $(Z_i, M_i) \perp\!\!\!\perp Y_i(z, m) \mid \mathbf{X}_i \rightsquigarrow \mathbb{E}\{Y_i(z, m) \mid \mathbf{X}_i\}$ is identifiable
- (C): $Z_i \perp\!\!\!\perp M_i(z) \mid \mathbf{X}_i \rightsquigarrow \mathbb{E}\{M_i(z) \mid \mathbf{X}_i\}$ is identifiable
- (A)+(B)+(C) hold under experiments with sequentially randomized treatment and mediator

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- (C): $Z_i \perp\!\!\!\perp M_i(z) \mid \mathbf{X}_i \rightsquigarrow \mathbb{E}\{M_i(z) \mid \mathbf{X}_i\}$ is identifiable
- (A)+(B)+(C) hold under experiments with sequentially randomized treatment and mediator
- (D) is fundamentally meta-physical because no physical experiment can ensure it.

# Mediation formula (Pearl, 2001)

**Theorem**

*Under (A) to (D),* $\mathbb{E}\{Y_i(z, M(z'))\} = \mathbb{E}\left[\mathbb{E}\{Y_i(z, M(z')) \mid \mathbf{X}_i\}\right]$*, where*

$$\mathbb{E}\{Y_i(z, M(z')) \mid \mathbf{X}\} = \sum_m \mathbb{E}(Y \mid Z = z, M = m, \mathbf{X}) \mathrm{pr}(M = m \mid Z = z', \mathbf{X}).$$

$$\begin{aligned}
& \mathbb{E}\{Y_i(z, M(z')) \mid \mathbf{X}\} \\
= \; & \sum_m \mathbb{E}\{Y_i(z, M(z')) \mid M(z') = m, \mathbf{X}\} \mathrm{pr}(M(z') = m \mid \mathbf{X}) \\
= \; & \sum_m \mathbb{E}\{Y_i(z, m) \mid M(z') = m, \mathbf{X}\} \mathrm{pr}(M(z') = m \mid \mathbf{X}) \\
= \; & \sum_m \mathbb{E}\{Y_i(z, m) \mid \mathbf{X}\} \mathrm{pr}(M(z') = m \mid \mathbf{X}) \\
= \; & \sum_m \mathbb{E}\{Y_i \mid Z = z, M = m, \mathbf{X}\} \mathrm{pr}(M = m \mid Z = z', \mathbf{X})
\end{aligned}$$

# Mediation formula

- Mediation formula for NDE($\mathbf{X}$) and NDE($\mathbf{X}$)

$$
\begin{aligned}
\mathrm{NDE}(\mathbf{X}) &= \mathbb{E}\{Y_i(1, M(0)) \mid \mathbf{X}\} - \mathbb{E}\{Y_i(0, M(0)) \mid \mathbf{X}\} \\
&= \sum_m \{\mathbb{E}(Y_i \mid Z = 1, M = m, \mathbf{X}) - \mathbb{E}(Y_i \mid Z = 0, M = m, \mathbf{X})\} \\
&\qquad \cdot \mathrm{pr}(M = m \mid Z = 0, \mathbf{X}) \\
\mathrm{NIE}(\mathbf{X}) &= \mathbb{E}\{Y_i(1, M(1)) \mid \mathbf{X}\} - \mathbb{E}\{Y_i(1, M(0)) \mid \mathbf{X}\} \\
&= \sum_m \mathbb{E}(Y_i \mid Z = 1, M = m, \mathbf{X}) \\
&\qquad \cdot \{\mathrm{pr}(M = m \mid Z = 1, \mathbf{X}) - \mathrm{pr}(M = m \mid Z = 0, \mathbf{X})\}
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&\qquad \cdot \text{pr}(M = m \mid Z = 0, \mathbf{X}) \\
\text{NIE}(\mathbf{X}) &= \mathbb{E}\{Y_i(1, M(1)) \mid \mathbf{X}\} - \mathbb{E}\{Y_i(1, M(0)) \mid \mathbf{X}\} \\
&= \sum_m \mathbb{E}(Y_i \mid Z = 1, M = m, \mathbf{X}) \\
&\qquad \cdot \{\text{pr}(M = m \mid Z = 1, \mathbf{X}) - \text{pr}(M = m \mid Z = 0, \mathbf{X})\}
\end{aligned}
$$

- Average over $\mathbf{X}$ to obtain the NDE and NIE

## Baron-Kenny method (Baron and Kenny, 1986)

- Mediation formula under linear models

$$
\begin{aligned}
\mathbb{E}(M \mid Z, X) &= \beta_0 + \beta_1 Z + \beta_X^\top X \\
\mathbb{E}(Y \mid Z, M, X) &= \theta_0 + \theta_1 Z + \theta_2 M + \theta_X^\top X
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{NDE}(\mathbf{x}) &= \sum_m [\mathbb{E}\{Y_i \mid Z = 1, M = m, \mathbf{X}\} - \mathbb{E}\{Y_i \mid Z = 0, M = m, \mathbf{X}\}] \\
&\qquad \cdot \mathrm{pr}(M = m \mid Z = 0, \mathbf{X}) \\
&= \sum_m \theta_1 \mathrm{pr}(M = m \mid Z = 0, \mathbf{X}) = \theta_1 \\
\mathrm{NIE}(\mathbf{x}) &= \sum_m \mathbb{E}\{Y_i \mid Z = 1, M = m, \mathbf{X}\} \\
&\qquad \cdot \{\mathrm{pr}(M = m \mid Z = 1, \mathbf{X}) - \mathrm{pr}(M = m \mid Z = 0, \mathbf{X})\} \\
&= \sum_m (\theta_0 + \theta_1 + \theta_2 m + \theta_X^\top X) \\
&\qquad \cdot \{\mathrm{pr}(M = m \mid Z = 1, \mathbf{X}) - \mathrm{pr}(M = m \mid Z = 0, \mathbf{X})\} \\
&= \theta_2 \{\mathbb{E}(M \mid Z = 1, \mathbf{X}) - \mathbb{E}(M \mid Z = 0, \mathbf{X})\} = \theta_2 \beta_1
\end{aligned}
$$

# Baron-Kenny method

$$
\begin{tikzcd}
 & X \arrow[dl] \arrow[d, "\beta_X"] \arrow[dr, "\theta_X"] & \\
Z \arrow[r, "\beta_1"] \arrow[rr, bend right, "\theta_1"'] & M \arrow[r, "\theta_2"] & Y
\end{tikzcd}
$$

- Regress $Y$ on $Z$, $M$ and $X \rightarrow \widehat{\theta}_1$ and $\widehat{\theta}_2$
- Regress $M$ on $Z$ and $X \rightarrow \widehat{\beta}_1$

- Point estimates: $\widehat{\text{NDE}} = \widehat{\theta}_1$ and $\widehat{\text{NIE}} = \widehat{\theta}_2 \widehat{\beta}_1$

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- Regress $Y$ on $Z$, $M$ and $X \to \widehat{\theta}_1$ and $\widehat{\theta}_2$
- Regress $M$ on $Z$ and $X \to \widehat{\beta}_1$

- Point estimates: $\widehat{\text{NDE}} = \widehat{\theta}_1$ and $\widehat{\text{NIE}} = \widehat{\theta}_2\widehat{\beta}_1$

- Variance of NIE: Delta method $\to$ the asymptotic variance of the NIE is $\text{var}(\widehat{\theta}_2)\beta_1^2 + \text{var}(\widehat{\beta}_1)\theta_2^2$
- Variance estimator $\widehat{\text{var}}(\widehat{\theta}_2)\widehat{\beta}_1^2 + \widehat{\text{var}}(\widehat{\beta}_1)\widehat{\theta}_2^2$

# With interaction

$$\begin{aligned}
\mathbb{E}(M \mid Z, X) &= \beta_0 + \beta_1 Z + \beta_X^\top X \\
\mathbb{E}(Y \mid Z, M, X) &= \theta_0 + \theta_1 Z + \theta_2 M + \theta_3 ZM + \theta_X^\top X
\end{aligned}$$

$$\begin{aligned}
\mathrm{NDE}(\mathbf{x}) &= \sum_m [\mathbb{E}\{Y_i \mid Z=1, M=m, \mathbf{X}\} - \mathbb{E}\{Y_i \mid Z=0, M=m, \mathbf{X}\}] \\
&\qquad \cdot \mathrm{pr}(M=m \mid Z=0, \mathbf{X}) \\
&= \sum_m (\theta_1 + \theta_3 m)\mathrm{pr}(M=m \mid Z=0, \mathbf{X}) = \theta_1 + \theta_3(\beta_0 + \beta_X^\top X) \\
\mathrm{NIE}(\mathbf{x}) &= \sum_m \mathbb{E}\{Y_i \mid Z=1, M=m, \mathbf{X}\} \\
&\qquad \cdot \{\mathrm{pr}(M=m \mid Z=1, \mathbf{X}) - \mathrm{pr}(M=m \mid Z=0, \mathbf{X})\} \\
&= \sum_m (\theta_0 + \theta_1 + \theta_2 m + \theta_3 m + \theta_X^\top X) \\
&\qquad \cdot \{\mathrm{pr}(M=m \mid Z=1, \mathbf{X}) - \mathrm{pr}(M=m \mid Z=0, \mathbf{X})\} \\
&= (\theta_2 + \theta_3)\{\mathbb{E}(M \mid Z=1, \mathbf{X}) - \mathbb{E}(M \mid Z=0, \mathbf{X})\} = (\theta_2 + \theta_3)\beta_1
\end{aligned}$$

# Logistic model for binary mediator

$$
\begin{aligned}
\operatorname{logit}\{\mathrm{pr}(M=1 \mid Z, X)\} &= \beta_0 + \beta_1 Z + \beta_X^\top X \\
\mathbb{E}(Y \mid Z, M, X) &= \theta_0 + \theta_1 Z + \theta_2 M + \theta_X^\top X
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\mathrm{NDE}(\mathbf{x}) &= \sum_m \theta_1 \mathrm{pr}(M=m \mid Z=0, \mathbf{X}) = \theta_1 \\
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&\qquad \cdot \{\mathrm{pr}(M=m \mid Z=1, \mathbf{X}) - \mathrm{pr}(M=m \mid Z=0, \mathbf{X})\} \\
&= \theta_2 \{\mathbb{E}(M \mid Z=1, \mathbf{X}) - \mathbb{E}(M \mid Z=0, \mathbf{X})\} \\
&= \theta_2 \left( \frac{e^{\beta_0+\beta_1+\beta_X^\top \mathbf{x}}}{1+e^{\beta_0+\beta_1+\beta_X^\top \mathbf{x}}} - \frac{e^{\beta_0+\beta_X^\top \mathbf{x}}}{1+e^{\beta_0+\beta_X^\top \mathbf{x}}} \right) \\
\mathrm{NIE} &= \theta_2 \mathbb{E} \left( \frac{e^{\beta_0+\beta_1+\beta_X^\top X}}{1+e^{\beta_0+\beta_1+\beta_X^\top X}} - \frac{e^{\beta_0+\beta_X^\top X}}{1+e^{\beta_0+\beta_X^\top X}} \right)
\end{aligned}
$$

# Mediation analysis for a job training program

- A randomized field experiment that investigates the efficacy of a job training intervention on unemployed workers. The program is designed to not only increase reemployment among the unemployed but also enhance the mental health of the job seekers.
- Treatment $Z$: indicator of encouragement; mediator $M$: job-search self-efficacy; $Y$: measure of depressive symptoms
- Covariates $X$: age, gender, baseline depressive measure, etc.

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- Baron-Kenny method
  - NDE: point est. $= -0.035$, s.e. $= 0.011$
  - NIE: point est. $= -0.011$, s.e. $= 0.009$

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  - NDE: point est. $= -0.035$, s.e. $= 0.011$
  - NIE: point est. $= -0.011$, s.e. $= 0.009$

- Noncompliance is present in the study

# Connection between principal stratification and mediation analysis

- In strata with $M(1) = M(0)$, the indirect effect is zero

$$
\begin{aligned}
& \mathbb{E}\{Y(1) - Y(0) \mid M(1) = M(0)\} \\
= \quad & \mathbb{E}\{Y(1, M(1)) - Y(0, M(0)) \mid M(1) = M(0)\} \\
= \quad & \mathbb{E}\{Y(1, M(1)) - Y(1, M(0)) \mid M(1) = M(0)\} \\
& + \mathbb{E}\{Y(1, M(0)) - Y(0, M(0)) \mid M(1) = M(0)\} \\
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\end{aligned}
$$

- Principal strata direct effect: $\mathbb{E}\{Y(1) - Y(0) \mid M(1) = M(0) = m\}$
- VanderWeele (2008) studies the relations between the principal causal effects and natural direct and indirect effects
- Forastiere et al. (2018) discuss the connections between the assumptions

# Connection between principal stratification and mediation analysis

- Principal strata indirect effect:
  $\mathbb{E}\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 0\}$ ?

$$
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$$
\begin{aligned}
& \mathbb{E}\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 0\} \\
= \ & \mathbb{E}\{Y(1, M(1)) - Y(0, M(0)) \mid M(1) = 1, M(0) = 0\} \\
= \ & \mathbb{E}\{Y(1, 1) - Y(0, 0) \mid M(1) = 1, M(0) = 0\}
\end{aligned}
$$

- $\mathbb{E}\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 0\}$ consists of both direct and indirect effects

# Summary

- Mediation analysis studies the extent to which an effect is mediated through a particular pathway and to which the effect of a treatment on the outcome operates directly

- Natural direct and indirect effects $\rightarrow$ definitions rely on nested potential outcomes

- Identification assumptions
  - no treatment-outcome confounding
  - no mediator-outcome confounding
  - no treatment-mediator confounding
  - cross-world independence

- Different mediation formula under different models

# Extensions

- Sensitivity analysis for mediation analysis (Imai et al., 2010)
- Partial identification without cross-world independence
- Multiple mediator
  - generalization of the mediation analysis to more than one mediators $\rightsquigarrow$ path analysis
  - focus on one mediator $M_i$ $\rightsquigarrow$ NIE is the effect through $M_i$ and NDE is the sum of the direct effect and the effect through other mediators

# Suggested readings

- Natural direct and indirect effects
  - Pearl. 2001. " Direct and indirect effects"
- Connection between principal stratification and mediation analysis
  - Forastiere et al. 2018. "Principal ignorability in mediation analysis: through and beyond sequential ignorability'
- Sensitivity analysis for mediation analysis
  - Imai et al. 2010. "Identification, inference and sensitivity analysis for causal mediation effects"